



METHODOLOGY AND THEORY

A comparative study of maintenance data classification based on neural networks, logistic regression and support vector machines

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Abstract

Purpose – The purpose of this paper is to compare the effectiveness of different analytical approaches, namely artificial neural networks, logistic regression and support vector machines to assess the health of a strainer located at the suction side of the pump.

Design/methodology/approach – Data used for simulation included exemplars from clean (represented by datasets after cleaning the suction strainer) and faulty conditions (represented by datasets prior to cleaning the suction strainer). The same datasets were used for modeling in order to compare how different techniques perform when fed with the same information.

Findings – Principal component analysis-based artificial neural networks proved to be better than other techniques in classifying maintenance datasets and predicting flow resistance from a clogged suction strainer.

Originality/value – The work highlights the comparative effectiveness of three predictive analytical techniques in classifying real plant data from a suction strainer. This will provide an opportunity for maintenance experts to see the effectiveness of different techniques as well as revealing valuable information about the relationship between the condition of the suction strainer and the overall performance of the pump.

Keywords Maintenance, Neural nets, Oil industry

Paper type Research paper



The authors would like to thank company engineers and experts for providing the data for analysis. The authors also specially thank scientific research staff, particularly David Siegel and Edzel Lapira of IMS Center, University of Cincinnati, USA, for their additional support in analytical work.

Introduction

Oil and gas (O&G) production processes are quite complex in nature, a fact that can affect systems/equipments in a number of ways. Hydrocarbons contain a mix of water, CO₂, H₂S, sand and mud etc. These are processed and separated into oil, gas and water streams where oil and gas are exported to onshore. Pumps are commonly used as a means to export oil. Impurities in the fluid can cause several operational problems in export pumps, leading to both planned and unplanned stoppages. To prevent such problems, different types of strainers/filters are used on the suction side (upstream side) of the pumps. Figure 1 shows a schematic diagram of the pump including placement of the suction strainer.

Suction strainers are normally installed with differential pressure monitoring that indicate the clogging state of the strainer. Strainer manufacturers recommend maximum allowable differential pressure limits for strainers indicating when the filters must be cleaned/replaced. Running pumps with a clogged strainer may lead to problems such as insufficient capacity, elevated vibrations or noise levels, insufficient suction/discharge pressure, cavitation, stall, and excessive power demands which have adverse effects on the overall pumping efficiency (Bloch and Geitner, 1999; Girdhar and Octo, 2005). Regular cleaning of suction strainers is, therefore, recommended as a standard preventive maintenance activity for pumps. In most facilities, the differential pressure trending is also available for real-time online analysis and equipment history.

Proper maintenance of oil export pumps is crucial in keeping an optimum production profile. Any planned or unplanned downtime of these pumps can impact the overall export capacity of the platform. In the facility under study, there are three parallel connected pumps dedicated to oil export. In normal operation, two pumps are running, whereas the third acts as a redundant unit. Two of these pumps are driven by variable speed motors, whereas the third has a fixed speed driven motor mechanism. Maintenance logs shows that in one year of continuous operation of a pump, 60 percent of pump total planned downtime corresponded to the cleaning of the clogged suction strainer. The differential pressure transmitter for a suction strainer has set values of 0 and 1 BarG. A value of 0 or near zero represents a completely clean strainer, whereas a value of 1 or near 1 represents a fully/partially blocked strainer. The task of cleaning the suction filter is performed based on differential pressure readings. In practical terms, there are no quantitative means of identifying the influence of a partially clogged strainer on the overall efficiency of the pump.

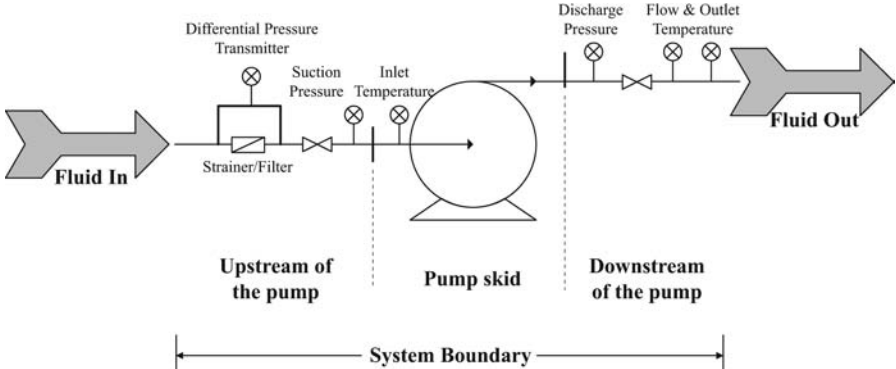


Figure 1.
Schematic diagram of
pumping operation

In this paper, we attempted to classify the maintenance data from both the clean and the partially filled state of the strainer. The collected data was labeled as “1”, representing healthy/clean state, and “0”, representing faulty/dirty state of the filter. We used three different techniques for data classification to see their applicability and efficiency in classifying the available data. These include logistic regression (LR), principal component analysis based artificial neural network (PCA-based ANN) and support vector machine (SVM) with linear discriminant function. The goal of classification models was to discriminate between the two classes, i.e. healthy or faulty states of the suction strainer. For modeling, one year of operational data from one of the variable speed driven centrifugal type oil export pumps was collected and analyzed. The ANN simulations were carried out at the Center for Industrial Asset Management (CIAM) at the University of Stavanger, Norway. Same data was analyzed by the experts at the Center for Intelligent Maintenance Systems (IMS) at the University of Cincinnati, USA, to test their tools from Watchdog Agent® Technology (www.imscenter.net). The data used for modeling was similar and included exemplars from clean data and faulty datasets. We did this so that we could compare how the models would perform when each analytic technique was fed with the same information.

Literature review

Predictive and statistical analytical techniques, such as rule-based expert systems, case-based reasoning and machine learning techniques have been extensively used to solve complex industrial problems (Schumacher *et al.*, 1996; McCormick and Nandi, 1997; Smith *et al.*, 2003; Xu *et al.*, 2005; Raza and Liyanage, 2009a, 2010). The purpose of employing such sophisticated techniques is to automatically extract knowledge from the data and to construct different model representations to explain complex phenomena. A bigger challenge lies in selecting the appropriate and efficient solution strategies in order to form a robust decision support (Raza and Liyanage, 2009a, b). According to Huang *et al.* (2004), the main difference between traditional statistical and machine learning methods is that statistical methods usually need researchers to impose structures to different models and to construct the models by estimating parameters to fit the data or observations that are relatively simple and easy to interpret. On the other hand, machine learning techniques allow learning the particular structure of the model directly from the data, but these models are usually very complicated and hard to explain. The application of machine learning techniques, in particular to the petroleum industry, is still a relatively unexplored area.

More and more predictive techniques are now being developed to assess and optimize the overall efficiency and performance of complex machineries. In general, pattern recognition tasks can be divided into two major types: namely, classification and estimation. Both of these are extensively applied in fault detection and diagnostics purposes. Classification deals with the process of applying mathematical and statistical techniques to separate data into different classes (i.e. diagnosing anomalies). These classes may differentiate normal and abnormal conditions of the processes. Estimation is the process of applying a potentially infinite set of numerical labels to an observation, such as estimating the health of a process (i.e. prognosis). These methods generally provide valuable information for automated fault detection and diagnosis. Fault diagnosis is a well developed area, whereas a lot of work needs to be done to improve the level of prognostics and decision-making (Principe *et al.*, 2000; Jamshidi, 2005).

With increasing demand for smarter diagnostic and prognostic solutions, Yan and Lee (2005) introduced an innovative approach based on Watchdog Agent[®] technology for performance degradation assessment and life prediction. This technology assesses and predicts equipment performance based on the inputs from the sensors mounted on it. Performance-related information is extracted through signal processing, feature extraction and fusion techniques. Historical patterns of the process signatures are utilized to predict the behavior and thus forecast the process or machine performance. The prognostic capabilities of Watchdog Agent[®] technology is based on tools for signal processing, performance prediction, health assessments and health diagnostics (Lee, 1996). Watchdog Agent[®] technology is a suite of several analytical software tools include time domain analysis, principal component analysis (PCA), fuzzy logics (FLS), logistic regression (LR), artificial neural network (ANN), Bayesian belief networks and support vector machines (SVM) etc. The combination of these tools is determined by the type of data and the desired objectives.

LR is a technique for analyzing problems where there are one or more independent variables that determine an outcome with a dichotomous variable, i.e. only two possible outcomes. This is to find the best fitting model to describe the relationship between the dichotomous characteristic of the dependent variable and a set of independent variables. Here, the LR is used to set up the relationship between clean and partially/fully blocked suction strainer condition. LR separates the two classes by an S-shaped discriminant function through the decision space (Agresti, 1996). If there is more than one independent variable, the discriminant is a plane called a separating hyperplane. The observations falling each side of the curve or hyperplane are classified as members of two different classes. The S-shaped discriminant function or hyperplane is regarded as a linear separator (Bishop, 1995).

An ANN system consists of processing elements (PEs) that are linked together. The basic building block of an ANN is a perceptron that is a weighted linear combination of independent variables plus a bias term in the input to the activation function. A multi-layer perceptron (MLP) is a feed forward ANN, consisting of input, hidden and output layers. MLPs are widely known for solving generalized problems (Narendra and Parthasarathy, 1990; Gonzalez and Desjardins, 2002). These networks have found their way into countless applications in data classification and pattern recognition tasks (Haykin, 1999; Principe *et al.*, 2000; Raza and Liyanage, 2009a). Correlation analysis of input data samples consisted of correlated information caused by overlapping input instances. This overlapping can create confusion in the training of ANN and therefore may reduce the network's generalized capabilities. To avoid such problems, we applied PCA technique prior to building the ANN so that the uncorrelated data is used to train a MLP neural network model (Mohamad-Saleh and Hoyle, 2008).

SVMs are a new classification method based on the statistical learning theory of Vapnik (1996). A SVM with linear classification function is a hyperplane that separates the two sets with a great margin. The margin is the sum of distances from the plane to the closest point of each set. Statistical learning theory proves that maximizing the margin of separation minimizes expected misclassification error on future unseen data. Maximizing the margin of separation prevents overfitting the training data even when the dimensionality of attributed space is very high. Originally, SVMs were used for pattern recognition, but now they are being extensively used for nonlinear regression estimation, time series prediction and classification, condition monitoring and fault

diagnostics. For further reading, see Mukherjee *et al.* (1997), Vapnik *et al.* (1997), Tay and Cao (2002), Yuan and Chu (2006), and Widodo and Yang (2007).

Briefly, the above-mentioned techniques fall into two main data classification approaches. The first considers only a dichotomous distinction between the two classes and assigns class labels, 0 or 1, to an unknown data item. The second yields not only a class label for a data item, but also a probability, P , of class membership. SVMs are the most prominent representatives of the first class, whereas LR and ANN are members of the second class (Dreiseitl and Lucila, 2002).

All of the above-mentioned techniques are well explored, and abundant literature is available, particularly on the industrial application of ANN, SVM and LR in various sectors. The cited literature showed that the comparative study has been attempted for pattern classification mostly in epidemiological research, medicine and clinical studies, and health care research (for instance see Freeman *et al.*, 2000; Gonzalez and Desjardins, 2002; Adeodato *et al.*, 2004; Song *et al.*, 2005; Xu *et al.*, 2005; Yuan and Chu, 2006; Widodo and Yang, 2007). We found scarcity of literature that attempts to compare the effectiveness of these techniques, particularly in relation to classifying maintenance data from heavy rotating equipment in the petroleum industry. The work presented in this paper compares the mutual effectiveness of these techniques to classify healthy and faulty datasets from centrifugal type oil export pumps. Such information can provide additional knowledge for planning and executing preventive maintenance tasks.

Technical description of datasets

For modeling purposes, one year of sensors' time series datasets from a centrifugal type oil export pump were used. Based on the maintenance logs, the data was labeled as healthy (1) and faulty (0). Datasets labeled "0" represent clogged strainer, whereas datasets labeled as "1" represent clean state of strainer. This resulted in a total of 8,383 exemplars in which 4,157 represented healthy strainer, whereas 4,225 represented filled/partially filled strainer condition.

Another dataset, comprising of 8,042 exemplars, was collected from the second similar pump including healthy (3,994) and faulty (4,048) exemplars. This dataset was not used during training or testing of the developed models and was used for testing generalized capabilities of neural networks.

The preliminary analysis of healthy and faulty data showed some significant features that revealed a relationship between the condition of the suction strainer and pump performance. Plots of differential pressure variations during clean and clogged suction strainer states are shown in Figures 2 and 3.

Correlation analysis of the raw data showed a high degree of correlation between strainer differential pressure and other parameters such as flow, current and pump head. Preliminary data analysis also discovered that higher differential pressure of the strainer may result in:

- higher flow resistance;
- reduced output; and
- higher loads on the motor power.

To classify these datasets, we attempted to model the data using different techniques. Discussion of the obtained results is given in the next sections of this paper.

Figure 2.
Clean state of suction
strainer

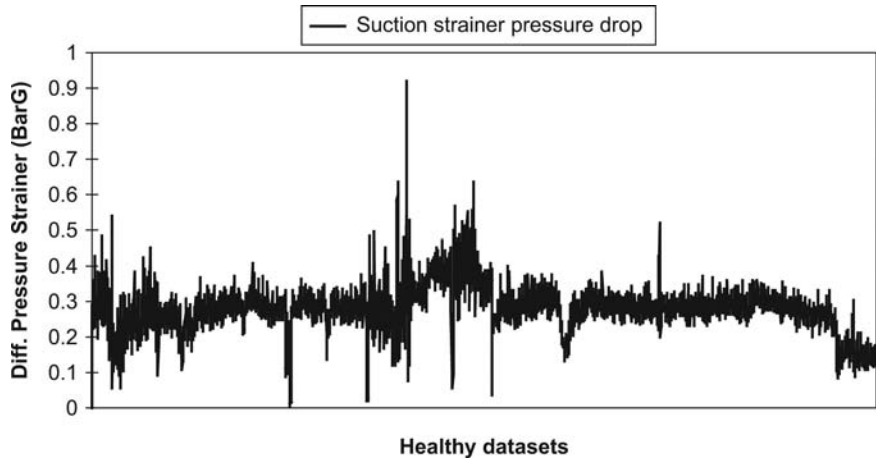
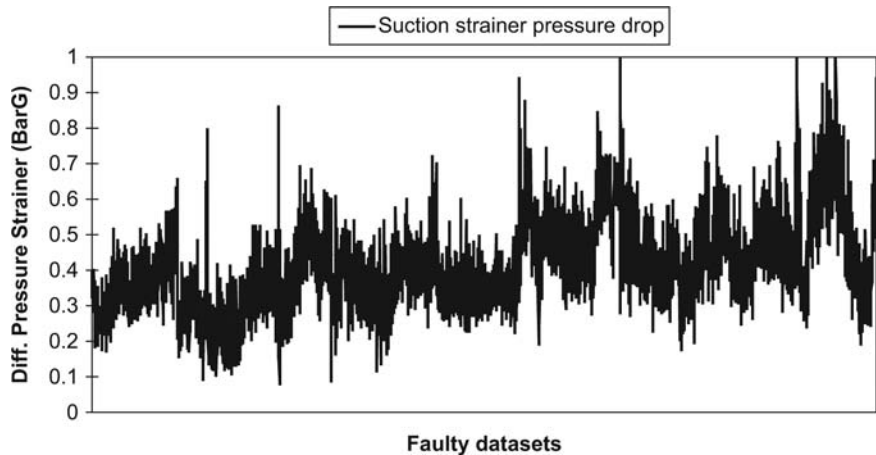


Figure 3.
Clogged state of suction
strainer



Results and discussions

ANN modeling results

The data for developing the ANN model was collected, processed and analyzed using commercially available neuro-computational software called NeuroSolutions 5.0. The data was pre-processed to remove outliers and idle operations. The data was then checked and cleaned from missing data/nuisance using inbuilt data pre-processing modules in NeuroSolutions 5.0 software. Correlation analysis of input variables was performed and some non-correlated variables were removed to simplify the modeling task. The final selected input variables for ANN modeling included inlet temperature, outlet pressure, flow, speed, current and motor power. The original data was randomized so that the representative data is included in all three sets including training (60 percent), cross validation (15 percent) and testing (25 percent). The network was trained using different algorithms in order to find the

most efficient classification results of these two datasets. These included models based on multilayer perceptron (MLP) network, generalized feed forward network (GFF) and principal component analysis (PCA)-based MLP network. Applying the PCA input selection to MLP ANN showed improved accuracy as shown by the results summarized in Table I.

PCA-based MLP networks combine unsupervised and supervised learning in the same topology. PCA is an unsupervised linear procedure that finds a set of uncorrelated features – principal components – from the input. A MLP is supervised to perform the nonlinear classification from these components. PCA-based ANNs are generally more efficient in classification tasks (Lerner *et al.*, 1999; Zhang, 2007; Mohamad-Saleh and Hoyle, 2008). This was found true in our analytical results. Data classification efficiency of all networks was calculated using the following error function:

$$Efficiency = \frac{Output_{real} - Output_{network}}{Output_{real}} \times 100 \quad (1)$$

Same data was analyzed using tools from Watchdog Agent[®] (www.imscenter.net) and the results are summarized in the following section.

Watchdog Agent[®] approach

The preliminary data analysis was conducted in accordance with the 5S methodology developed by the IMS Center. Tools from the Watchdog Agent[®] prognostic platform were utilized to estimate the health information of the pump under investigation. An overview of the 5S methodology is given in Figure 4 (Djurdjanovic *et al.* 2004).

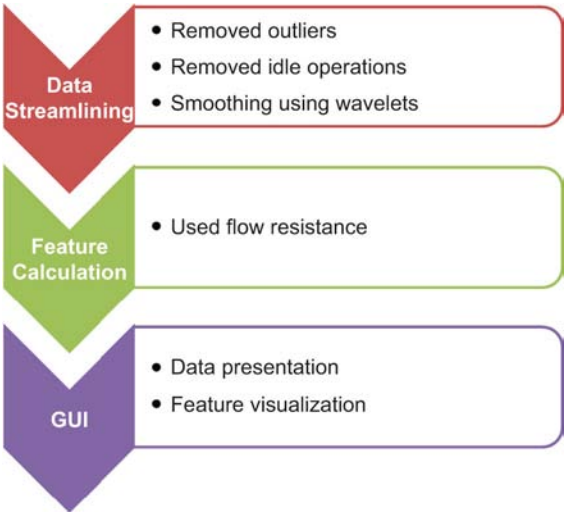
The data were streamlined to remove outliers and idle operations. Additional smoothing using wavelets was applied to remove noise and nuisance from the data. In order to better understand the overall signal trend and performance of the health assessment method, the fluid flow and pressure drop signal were de-noised. This removed all high frequency noise and disturbances in the raw pressure and fluid flow signals. Features were extracted from the data that showed a flow resistance feature (R) was very representative of the strainer health. According to Bernoulli's principle, pressure drop across a filter is proportional to the fluid flow squared and using this concept, a resistance feature can be given as the pressure drop divided by the fluid flow rate squared (Borutzky *et al.*, 2002):

$$R_{Feature} = \frac{\Delta P_{Filter}}{(Flow)^2} \quad (2)$$

Models	Network performance		Generalization	
	Healthy exemplar (pump 1) (%)	Faulty exemplar (pump 1) (%)	Healthy exemplar (pump 2) (%)	Faulty exemplar (pump 2) (%)
MLP	3	3.8	53	28
GFF	15	15	19	10
PCA-based MLP	0.4	0.39	14	7

Table I.
Data classification
efficiency for different
ANN algorithms

Figure 4.
Methodology for
Watchdog Agent®
prognostic platform



A plot of the resistance features for the healthy and faulty data is shown in Figure 5, which showed that there is a much higher level of resistance to the flow prior to the strainer cleaning.

As aforementioned, higher pressure indicates an increased level of clogging of the strainer and add up to higher flow resistance. This feature was much less in the clean state of the suction strainer. For this particular event, since the feature was very stable along with the physical intuition of this flow system, it was sufficient to use Logistic

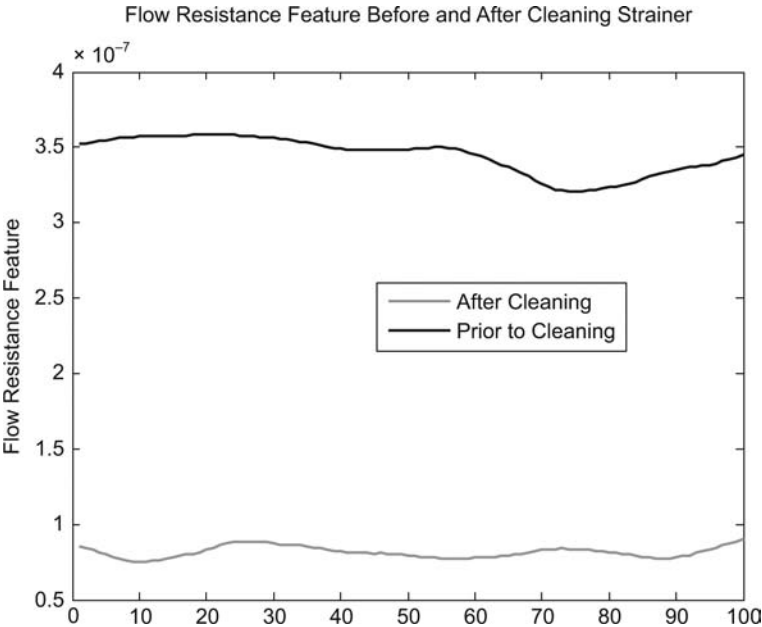


Figure 5.
Resistance feature plot for
clean and clogged strainer
conditions

Regression (LR) for training the health assessment algorithm. This flow resistance was selected while classifying data for LR application. Figure 6 shows how healthy and faulty datasets can be labeled based on this flow feature.

The use of LR was very suitable for this situation since both healthy and degraded datasets were available for training. The desired output is a health indicator value for the strainer cleaning event. In this particular situation, the dependent variable was the strainer clogging event, while the independent variable was the flow resistance feature. Mathematically, the logistic function is shown below in equations (3) and (4):

$$P(\mathbf{x}) = \frac{1}{1 + e^{-g(\mathbf{x})}} = \frac{e^{g(\mathbf{x})}}{1 + e^{g(\mathbf{x})}} \quad (3)$$

$$\text{Logit} = g(\mathbf{x}) = \log \left(\frac{P(\mathbf{x})}{1 - P(\mathbf{x})} \right) = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k \quad (4)$$

The training process involved fitting the parameters α and $\beta_1 \dots \beta_k$, to the training dataset. Once the logistic model was trained, the incoming data and extracted features can be input into the trained model and the output represents health state at that given instance in time (Yan and Lee, 2005). However, this particular strainer condition monitoring required only one feature. The testing set consisted of the pressure drop and flow data for the pump containing healthy and faulty datasets. This testing set was used to validate this method and see whether it could show the degradation of this strainer problem over time up until the strainer needed cleaning. A plot of the resistance feature for the testing set of data in Figure 7 showed an increase trend over time.

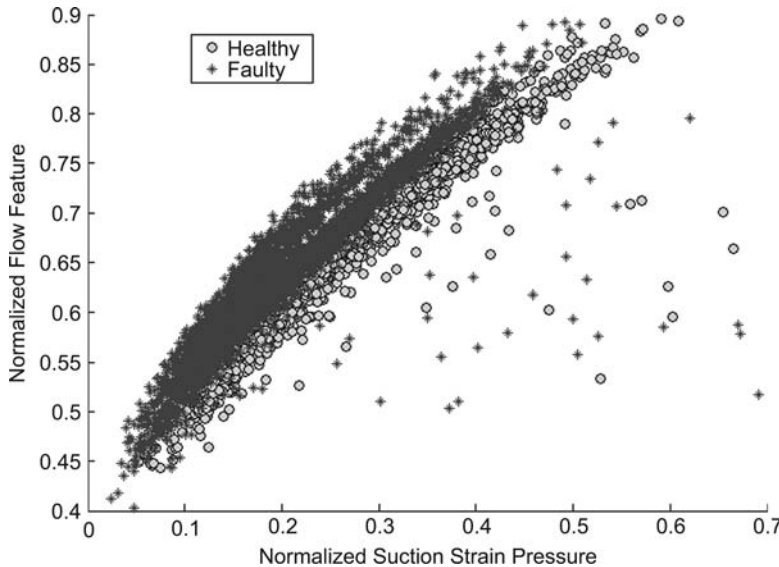


Figure 6.
Healthy and faulty data
labeling

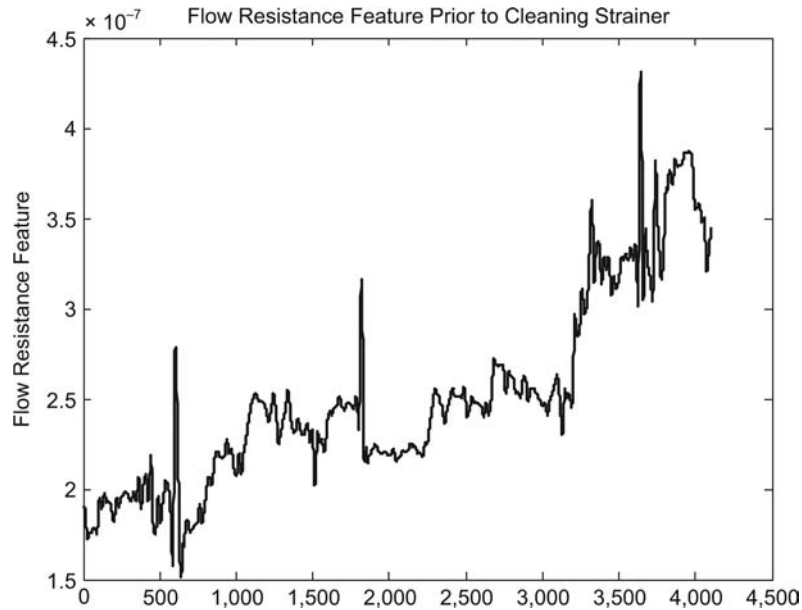


Figure 7.
Flow resistance feature
prior to cleaning strainer

A health indicator value for this event was developed and a predictive maintenance approach was used to predict strainer health state based on the trend. This monotonically increasing trend kept increasing with time until the strainer became clogged. Using the LR technique, the calculated health indicator value indicates the level of degradation. The health indicator value is given a value between 0 and 1, where the values closer to 0 indicate a higher level of degradation or clogging of the strainer. For this dataset, the health indicator value was calculated and a plot of the results showing the trend prior to cleaning is given in Figure 8.

The state closer to 1 represents an initially clean strainer whose cleanness reduces with time and eventually approaches 0, representing the state of clogged strainer. This health monitoring approach can be used for more predictive maintenance as well as for deciding how best to allocate resources or determine the schedule of use for the pump. For example, if it is known from the health indicator value that the strainer needs to be cleaned in two weeks, then the necessary resources and personnel can be scheduled for this maintenance action in advance.

To see the applicability of other suitable tools, the data was further analyzed using support vector machine (SVM) with linear function in order to best classify the two datasets. The preliminary results showed that the health of the strainer can be accurately estimated using both LR and SVM methods. Figure 9 shows classification results while using SVM technique.

To compare the classification efficiency of LR and SVM models, error function was calculated based on equation (1) (Table II).

The results obtained from the analysis can be used to limit any unforeseen downtime regarding the pump system, since such a health monitoring system can provide the information regarding the degradation and degradation levels. This preliminary study can be seen as a start to developing a robust health monitoring

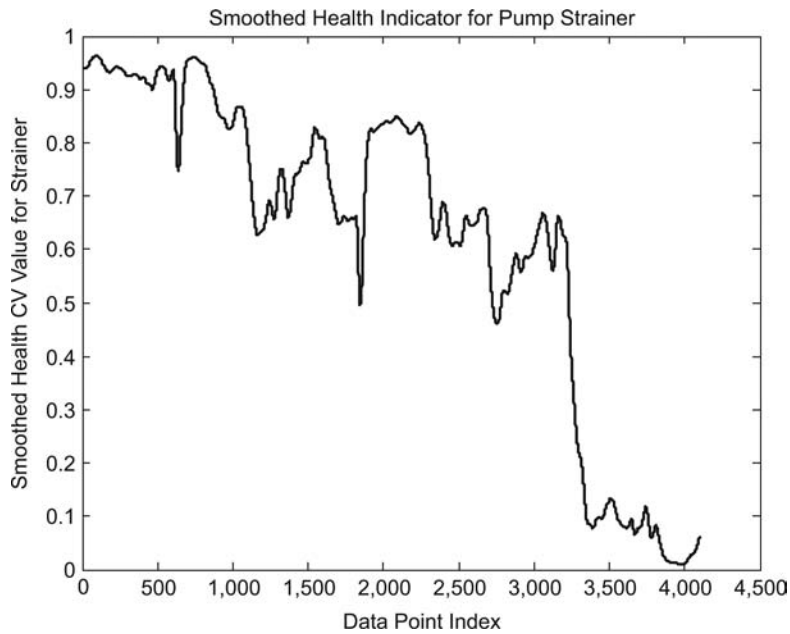


Figure 8.
CV value for pump B
strainer

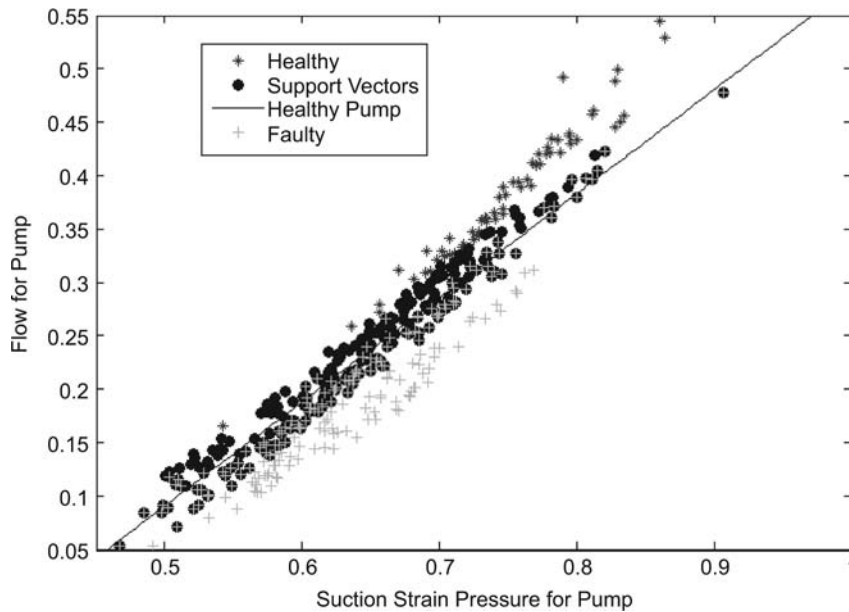


Figure 9.
Support vector machine
(SVM) for data
classification

system for the pump systems, using the experience and health monitoring tools developed at the IMS Center.

Conclusions

In this paper, we compared different analytical techniques for data classification from a suction strainer installed on the inlet of oil export pumps. There are potentially large numbers of mathematical and statistical approaches for performing such a task. LR and SVM techniques were selected from the Watchdog Agent[®] prognostic platform to analyze and assess the overall health of the strainer/pump system. Similar datasets were analyzed using ANN.

ANNs have many advantages over LR and SVM as they can learn from examples with a small number of prior assumptions about structural relationships. They proved to be more useful to fully recognize the complexities of the data when compared to LR technique. ANNs work well in pattern recognition tasks incorporating missing data and incurring problems of multicollinearity. However, fundamentally, both LR and ANN are different approaches. ANNs also have the ability to model highly nonlinear functions that can be trained accurately and generalized when presented with new, unseen datasets. These features make them attractive to developing numerical models as compared with other statistical approaches. However, they have some inherent problems that can limit their further applications such as determination of network architecture, estimation of suitable learning parameters, selection of training data, local minima, slow learning speed etc.

SVM is a valuable pattern recognition method in theory and in application. Compared with other algorithms, it is faster with a higher precision ratio. However, conventional SVM employs the structural minimization principle; thus, empirical risk of misclassification may be high, especially when a point to be classified is close to the hyperplane.

We compared different approaches in analyzing the suction strainer condition in relation to its effect on the pump. We outlined common foundations for data classification modeling and compared their individual efficiencies through obtained results. In conclusion, we found neural networks to be superior to both LR and SVM based on the simulated results. However, LR proved to be quick to simulate, whereas ANN and SVM both were clock-intensive.

Data classification results from the three techniques being employed are summarized in Table III, showing ANN as a clear winner for this specific problem.

In our analysis, ANN showed better classification results as compared to the other two techniques. ANN also produced satisfactory results when tested for data from another similar unit. Figures 10 and 11 show flow predictions for clean and clogged strainer conditions.

Table II.
Classification results
from SVM and LR models

Tools	Network performance	
	Healthy exemplar (pump 1) (%)	Faulty exemplar (pump 1) (%)
SVM	6.86	3.63
LR	2.41	1.28

As expected, ANN proved to be quite efficient in predicting flow resistance in both clean and clogged strainer conditions. Such an approach can potentially be useful for planning and scheduling maintenance tasks. Moreover, it revealed a hidden relationship between filter differential pressure and performance parameters of the pump.

Effective utilization of generated information through the use of advanced diagnostic and prognostic tools can directly influence decision making and

Models	Network performance	
	Healthy exemplar (pump 1) (%)	Faulty exemplar (pump 1) (%)
SVM	6.86	3.63
LR	2.41	1.28
PCA-based MLP	0.4	0.39

Table III.
Comparative data
classification results

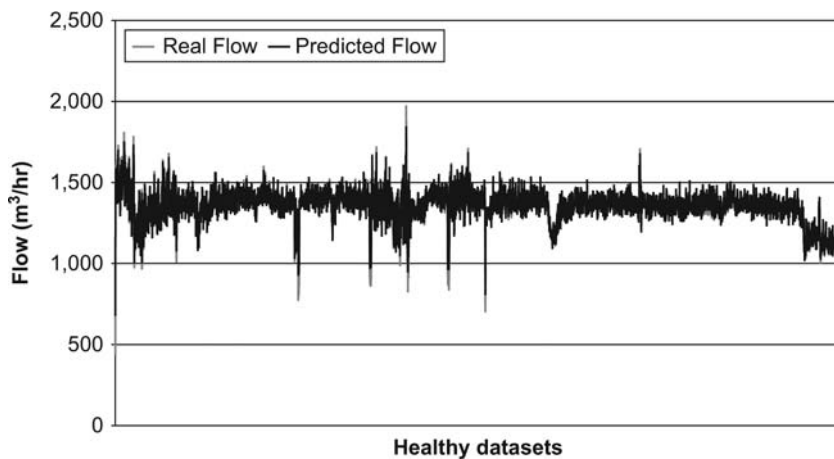


Figure 10.
Predicted flow for clean
strainer

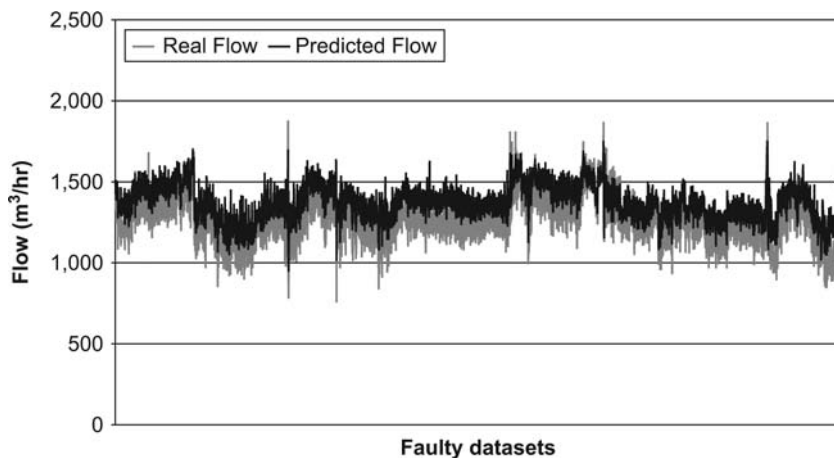


Figure 11.
Predicted flow for clogged
strainer

cost-effective maintenance planning. This will allow a better and more visible monitoring of critical components of a system thereby revealing complex relationship among different variables that can be very helpful in identifying hidden system and process faults/anomalies prior to their occurrence. Such tools can work either independently or as an integral part of existing analytical tools. Practical implications of such useful information can lead to manual generation of corrective work orders in computerized maintenance management system (CMMS) that can result in huge cost savings in terms of on-time maintenance rather than calendar-based. These techniques can be effectively employed with slight interface modifications to visualize online performance of a system. Furthermore warnings can be associated with these alerting the discipline engineer(s)/maintenance managers in form of automated notification (SMS, e-mails, pagers). This preliminary study can be seen as a start to developing a robust health monitoring system for the pump systems using the experience and health monitoring tools jointly employed by the CIAM and the IMS Center.

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